OPTIMAL CONDITIONING AND LOCAL PERTURBATION CRITERIA FOR ROW-WEIGHTED VANDERMONDE MATRICES ON THE UNIT CIRCLE

Abstract. For square Vandermonde matrices on the unit circle with fixed nonzero monomial row weights, the minimum spectral condition number is the ratio of the largest to the smallest weight modulus, and every minimizer is a rotated regular polygon. At each fixed dimension, a Fourier vanishing condition on the extreme-weight index sets characterizes perturbation directions with no first-order increase near a minimizer. For unequal weight moduli, every such non-rotational direction has strictly quadratic growth, with a computable coefficient given by two effective spectral matrices and explicit positive lower bounds. A weighted Fourier moment constraint follows by optimizing coupled spectral bounds. For triangular weights arising from geometric-square representations of binary forms, flatness reduces to vanishing of the first discrete Fourier coefficient, and the second-harmonic coefficient is obtained in closed form. High-precision checks illustrate the theory, and Fourier-based computations are compared with precomputed dense and cached LU methods.

Key words. Vandermonde matrix; spectral condition number; optimal nodes; eigenvalue perturbation; discrete Fourier transform; binary form

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1. Introduction and problem setting. The problem studied here is to minimize $\kappa_2(WV)$ over distinct unit-circle nodes. The matrix V has entries $V_{n,k} = \beta_k^n$, and $W = \mathrm{diag}(w_0, \ldots, w_{L-1})$ contains fixed nonzero monomial row weights. The optimal value is $\max_n |w_n| / \min_n |w_n|$. The main results characterize all minimizing nodes and classify local perturbation directions by the growth order of this condition number. In particular, repeated extreme weights determine which Fourier components of a direction must vanish to avoid a linear increase.

The perfect conditioning of the unweighted Fourier matrix is classical. Berman and Feuer [3, Lemma 3.1] characterize perfect conditioning within a parametrized unit-circle family, while Pan [9] discusses the relation between node distributions and ill-conditioning. The closest comparison is Kuian, Reichel and Shiyanovskii [8]: their Lemma 5 gives orthogonality for equally spaced nodes, Theorem 8 permits scaling the rows indexed by nodes to equalize singular values, and Theorem 10 studies general polynomial bases on a prescribed equally spaced grid. In their transposed convention, our matrix is $V^T W$: its weights act on the monomial columns and are not optimization variables. Thus Theorem 3.1 concerns all minimizers under fixed monomial weights, with an optimum that need not be 1; the local Fourier criterion in Theorem 3.6 concerns how perturbations leave those minimizers.

The spectral perturbation tools themselves are standard. Chen, Womersley and Ye [5, Sections 2 and 4] analyze nonsmooth Gram-matrix condition numbers and optimize nodes and sampling weights. Standard Hermitian perturbation theory gives abstract compressed-matrix expansions. The Vandermonde structure here turns those compressions into explicit Fourier vanishing conditions on the extreme-weight index sets; the moment constraint and effective matrices then establish strictly positive quadratic growth in every remaining non-rotational flat direction. Proposition 3.4 derives a weighted Fourier moment bound by optimizing a scalar relaxation of the coupled spectral constraints, using the Bhatia–Davis variance bound [4]. Large-sieve bounds such as those of Aubel and Bölcskei [1] instead control singular values through node geometry. The discrete

Kadec estimate of Yu and Townsend [13] is used only as an existing finite-perturbation bound. Corollary 3.7 and Proposition 3.8 specialize the local theory to triangular weights, giving a first-harmonic criterion and an explicit quadratic coefficient.

The motivating application is a fixed-node representation of binary forms. Let $d \geq 1$, $m = 2d$ and $L = m + 1$, and write

$$F(x, y) = \sum_{n=0}^{m} A_n x^{m-n} y^n \in \mathbb{C}[x, y]_m, \qquad A = (A_0, \ldots, A_m)^T. \tag{1.1}$$

For the geometric form

$$Q_\beta(x, y) = \sum_{j=0}^{d} \beta^j x^{d-j} y^j, \tag{1.2}$$

the desired representation is

$$F(x, y) = \sum_{k=0}^{L-1} \lambda_k Q_{\beta_k}(x, y)^2. \tag{1.3}$$

The nodes are shared by all forms of the same degree, and zero representation coefficients λ_k are allowed. The fixed-node coefficient transformation studied here is distinct from classical Waring decomposition, which seeks representations with a minimal number of terms; see Bender et al. [2] and Reznick [10]. All condition numbers use the standard coefficient norm and the normalization in (1.2).

The main contribution is the local classification in Theorem 3.6: a Fourier necessary and sufficient condition determined by the extreme-weight index sets, together with a strictly positive quadratic lower bound for every non-rotational flat direction when the weight moduli are unequal. The same theorem gives its quadratic coefficient through the extreme effective matrices, with lower bounds expressed in terms of weight gaps and Fourier coefficients. Proposition 3.8 further derives an explicit second-order coefficient for the second-harmonic perturbation direction. The all-minimizer result in Theorem 3.1 provides the global starting point for this local analysis.

Section 2 constructs the square basis; Section 3 develops the optimal-node and local perturbation theory. Sections 4 and 5 give the standard Fourier formulas, error estimates and real signed-square construction. Section 6 reports the numerical checks, and Section 7 concludes.

2. The square basis and explicit weight formulas. Let

$$N_n = N(n, d) = \begin{cases} n + 1, & 0 \leq n \leq d, \\ m - n + 1, & d < n \leq m, \end{cases} \qquad D = \operatorname{diag}(N_0, \ldots, N_m). \tag{2.1}$$

By counting the integer pairs (i, j) with $i + j = n$ and $0 \leq i, j \leq d$, we have

$$Q_\beta(x, y)^2 = \sum_{n=0}^{m} N_n \beta^n x^{m-n} y^n. \tag{2.2}$$

Hence, if $V = (\beta_k^n)_{0 \leq n,k \leq m}$ and $M = DV$, then (1.3) is equivalent to

$$M\lambda = A, \qquad V\lambda = a, \qquad a = D^{-1}A. \tag{2.3}$$

THEOREM 2.1 (Fixed-node basis and dual formulas). *Suppose that $\beta_0, \dots, \beta_m$ are pairwise distinct. Then the squares $Q_{\beta_k}^2$ form a basis of $\mathbb{C}[x,y]_m$. If*

$$\ell_k(t) = \prod_{j \neq k} \frac{t - \beta_j}{\beta_k - \beta_j} = \sum_{n=0}^{m} w_{k,n} t^n,$$

then the unique decomposition weights are

$$\lambda_k = \sum_{n=0}^{m} w_{k,n} \frac{A_n}{N_n}, \tag{2.4}$$

and the associated differential dual operators are

$$\mathcal{L}_k = \sum_{n=0}^{m} \frac{w_{k,n}}{N_n (m-n)! n!} \partial_x^{m-n} \partial_y^n, \qquad \mathcal{L}_k(Q_{\beta_j}^2) = \delta_{kj}. \tag{2.5}$$

Proof. Since $N_n \neq 0$, the Vandermonde determinant formula gives

$$\det M = \prod_{n=0}^{m} N_n \prod_{0 \leq i < j \leq m} (\beta_j - \beta_i) \neq 0.$$

Therefore the squares are linearly independent and their number equals the dimension of the space. Substituting $a_n = \sum_j \lambda_j \beta_j^n$ into the right-hand side of (2.4) yields $\sum_j \lambda_j \ell_k(\beta_j) = \lambda_k$. Finally, (2.5) follows directly from the factorial factors in the homogeneous differential pairing. ∎

Define further the invertible diagonal transformation

$$\Phi_d(F) = \sum_{n=0}^{m} \binom{m}{n} \frac{A_n}{N_n} x^{m-n} y^n. \tag{2.6}$$

The square expansion gives $\Phi_d(Q_\beta^2) = (x + \beta y)^m$. This correspondence links the fixed geometric-square representation with the classical representation by powers of linear forms; the subsequent analysis keeps the triangular weights D in the original coefficient coordinates.

3. Optimal nodes and near-optimal constraints under general row weights. For an invertible matrix T write $\kappa_2(T) = \sigma_{\max}(T)/\sigma_{\min}(T)$. In this section let $L \geq 2$ be an arbitrary integer, $V = (\beta_k^n)_{0 \leq n,k < L}$, with distinct nodes satisfying $|\beta_k| = 1$, and fix

$$W = \operatorname{diag}(w_0, \dots, w_{L-1}), \quad w_n \neq 0, \qquad a = \min_n |w_n|, \quad b = \max_n |w_n|, \quad \rho = b/a.$$

The scalar a is used only in this section and is unrelated to the normalized coefficient vector of Section 2. The phases of w_n contribute only a unitary diagonal factor on the left; all singular values therefore depend only on the moduli $|w_n|$.

3.1. The optimal value and all optimal nodes.

THEOREM 3.1 (Weighted optimal condition number and all optimal nodes). *One has $\kappa_2(WV) \geq \rho$, and equality holds if and only if the nodes form a rotated regular L-gon, that is, there exist a real number θ and a permutation π such that*

$$\beta_{\pi(k)} = \mathrm{e}^{\mathrm{i}\theta} \mathrm{e}^{2\pi \mathrm{i} k/L}, \qquad 0 \leq k < L. \tag{3.1}$$

For these nodes the singular values are the multiset $\{\sqrt{L}|w_n| : 0 \leq n < L\}$.

Proof. Let $B = (WV)(WV)^*$. Then $B_{nn} = L|w_n|^2$, and the Rayleigh quotient gives

$$\lambda_{\max}(B) \geq Lb^2, \qquad \lambda_{\min}(B) \leq La^2, \tag{3.2}$$

whence $\kappa_2(WV) \geq \rho$. Under (3.1) we have $VV^* = LI$, so $B = LWW^*$, which yields the equality case and the singular value formula.

Conversely, suppose equality holds. Since $\lambda_{\max}(B)/(Lb^2) \geq 1$ and $La^2/\lambda_{\min}(B) \geq 1$, their product can equal 1 only if both inequalities in (3.2) are equalities. Choose any index j with $|w_j| = a$. Its Rayleigh quotient attains the minimum eigenvalue, so $Be_j = La^2 e_j$, and consequently

$$0 = B_{nj} = w_n \overline{w_j} \sum_{k=0}^{L-1} \beta_k^{n-j} \quad (n \neq j).$$

Write $p_r = \sum_k \beta_k^r$. Since $p_{-r} = \overline{p_r}$, we obtain

$$p_1 = \cdots = p_h = 0, \qquad h = \max(j, L-1-j) \geq \lfloor L/2 \rfloor.$$

Let e_r denote the elementary symmetric polynomials of the nodes, with $e_0 = 1$. The Newton identities

$$re_r = \sum_{q=1}^{r} (-1)^{q-1} e_{r-q} p_q$$

give $e_1 = \cdots = e_h = 0$, and the unit-circle condition gives

$$e_{L-r} = e_L \overline{e_r} \quad (0 \leq r \leq L), \qquad |e_L| = 1.$$

The reciprocal identity follows by replacing each product over $L-r$ nodes by e_L times the product of the reciprocals of the complementary r nodes. Since $2h \geq L-1$, the ranges $1 \leq r \leq h$ and $L-h \leq r \leq L-1$ cover all of $1, \ldots, L-1$, all these e_r vanish. The node polynomial is therefore $t^L - c$ with $|c| = 1$, whose roots are exactly (3.1). The choice of j was arbitrary: neither uniqueness of the extreme weights nor an endpoint minimum was used. When $a = b$, equality also gives $B = La^2 I$ and the same argument applies. ∎

Corollary 3.2 (The geometric square and higher powers). *For $L = 2d+1$ and $W = D$ of Section 2,*

$$\kappa_2(M) \geq d+1, \tag{3.3}$$

and the equality nodes are exactly (3.1)*. More generally, let $q \geq 1$ be an integer and let*

$$H(x,y) = \sum_{n=0}^{q} c_n x^{q-n} y^n, \qquad c_n \neq 0 \ (0 \leq n \leq q).$$

Then for $q+1$ distinct nodes the forms $H(x, \beta_k y)$ form a basis of $\mathbb{C}[x,y]_q$, and the optimal condition number on the unit circle is

$$\frac{\max_{0 \leq n \leq q} |c_n|}{\min_{0 \leq n \leq q} |c_n|},$$

with the same optimal configurations. In particular, for every integer $r \geq 1$, the coefficients of Q_β^r are the nonzero convolution coefficients of $(1 + t + \cdots + t^d)^r$, and $q = rd$.

Proof. The coefficient matrices are DV and $\mathrm{diag}(c_n)V$, so Theorem 3.1 applies. The coefficients of Q_β^r are positive integers. If some c_n of a general H vanishes, the corresponding matrix has a zero row and the full-space basis statement does not apply. ∎

COROLLARY 3.3 (Singular values and normalization). *At the optimal nodes of the geometric-square problem, the singular values of M are $\{\sqrt{L}N_n : 0 \le n \le 2d\}$. In particular,*

$$\|M\|_2 = \sqrt{L}(d+1), \qquad \left\|M^{-1}\right\|_2 = 1/\sqrt{L}.$$

After the input is normalized to $D^{-1}A$, the system matrix V has condition number 1.

3.2. A moment constraint for near-optimal configurations. We first record the standard spectral variance estimate. For a Hermitian matrix B, a unit vector x and $\ell I \le B \le uI$, with $\mu = x^*Bx$,

$$x^*B^2x - \mu^2 \le (\mu - \ell)(u - \mu). \tag{3.4}$$

This is the spectral form of the Bhatia–Davis variance bound [4]: in an orthonormal eigenbasis of B, the probability weights are the squared moduli of the components of x. The form for positive linear maps appears in Sharma and Thakur [12, eq. (1.2)], and the map $T \mapsto x^*Tx$ yields (3.4); it also follows directly from $(B - \ell I)(uI - B) \succeq 0$.

PROPOSITION 3.4 (Fourier moment constraint for near-optimal condition numbers). *Under the hypotheses of Theorem 3.1, define*

$$s_r = \frac{1}{L}\sum_{k=0}^{L-1} \beta_k^r, \qquad \eta = \frac{\kappa_2(WV)^2}{\rho^2} - 1 \ge 0.$$

Then for every minimum-weight row j,

$$T_j := \sum_{n \ne j} |w_n|^2 |s_{n-j}|^2 \le H_{a,b}(\eta), \qquad H_{a,b}(\eta) = \begin{cases} \dfrac{(b^2-a^2)\eta}{1+\eta}, & 0 \le \eta \le 1 - a^2/b^2, \\ \dfrac{[b^2(1+\eta) - a^2]^2}{4b^2(1+\eta)}, & \eta \ge 1 - a^2/b^2. \end{cases} \tag{3.5}$$

In particular, for the geometric-square weights with $j = 0$,

$$\sum_{n=1}^{2d} N_n^2 |s_n|^2 \le \frac{d(d+2)\eta}{1+\eta}, \qquad 0 \le \eta \le 1 - (d+1)^{-2}, \tag{3.6}$$

where $\eta = \kappa_2(M)^2/(d+1)^2 - 1$.

Proof. Write $\ell = \lambda_{\min}(B)$ and $u = \lambda_{\max}(B)$. Since $u/\ell = (b/a)^2(1+\eta)$, (3.2) implies $x := \ell/(La^2) \in [1/(1+\eta), 1]$. Applying (3.4) to e_j and using $B_{nj} = Lw_n\overline{w_j}s_{n-j}$ gives

$$T_j \le \frac{(La^2 - \ell)(u - La^2)}{L^2a^2} = (1-x)[b^2(1+\eta)x - a^2] =: g(x).$$

This concave quadratic has vertex $x_* = [b^2(1+\eta) + a^2]/[2b^2(1+\eta)]$. Its maximum on the stated interval is at $x = 1/(1+\eta)$ when $\eta \le 1 - a^2/b^2$, and at x_* otherwise. The two values are precisely (3.5). ∎

The upper bound is obtained by maximizing over the admissible scalar interval and need not be sharp. For fixed $b > a$ and $\eta \to 0$,

$$H_{a,b}(\eta) = (b^2 - a^2)\eta + O(\eta^2),$$

so $\sqrt{T_j} = O(\sqrt{\eta})$. When $a = b$, the identity $H_{a,a}(\eta) = a^2\eta^2/[4(1+\eta)]$ gives $\sqrt{T_j} = O(\eta)$. These estimates impose necessary Fourier moment constraints on near-optimal configurations, but do not directly characterize geometric node displacements or the exact growth order of local perturbations.

3.3. A Fourier criterion for the growth order of perturbation directions. The moment constraint is only necessary. We now fix a perturbation direction near the regular polygon and characterize the directions with only second-order growth. Let $h = (h_0, \dots, h_{L-1}) \in \mathbb{R}^L$ be fixed and set

$$\beta_k(t) = \mathrm{e}^{\mathrm{i}(\theta + 2\pi k/L + t h_k)}, \qquad \widehat{h}_r = \frac{1}{L}\sum_{k=0}^{L-1} h_k \mathrm{e}^{2\pi \mathrm{i} r k/L}. \tag{3.7}$$

Write $V(t) = (\beta_k(t)^n)$ and $J_- = \{n : |w_n| = a\}$, $J_+ = \{n : |w_n| = b\}$. For any nonempty index set J, define the Hermitian matrix

$$H_J(h) = \left[\mathrm{i}(n-m)\widehat{h}_{n-m}\right]_{n,m\in J}.$$

Its diagonal entries vanish, and a one-element index set gives the zero matrix. The constants in the expansions below may depend on the fixed L, W and h.

LEMMA 3.5 (Extreme spectral clusters). *Let $A(t) = A_0 + tE + O(t^2)$ be Hermitian in a fixed finite dimension, with the remainder bounded in operator norm, and let P project onto the top eigenspace of A_0, with eigenvalue μ. For $t > 0$,*

$$\lambda_{\max}(A(t)) = \mu + t\lambda_{\max}(PEP|_{\operatorname{ran} P}) + O(t^2).$$

The analogous minimum formula holds on the bottom eigenspace. For $t < 0$ the extremes of the compressed matrices are interchanged. These statements include the case $A_0 = \mu I$. If $P \neq I$, put $g = \mu - \lambda_{\max}(A_0|_{\operatorname{ran}(I-P)}) > 0$ and $e = \|E\|_2$. If $\|A(t) - A_0 - tE\|_2 \le Ct^2$ for $|t| \le t_0$, the absolute error in the displayed top-eigenvalue formula is at most $(C + 2e^2/g)t^2$ whenever $0 < t \le t_0$ and $4te \le g$.

Proof. Suppose the complementary space is nonempty, with spectral gap $g > 0$. Write a unit vector as $x = u + v$, where $u = Px$, and put $z = \|v\|_2$. Relative to $\mu + t\lambda_{\max}(PEP)$, its Rayleigh quotient is at most $-gz^2 + 2t\|E\|_2 z + 2t\|E\|_2 z^2 + Ct^2$ for a remainder bound C. For $2t\|E\|_2 \le g/2$, maximizing over z bounds this by $(C + 2\|E\|_2^2/g)t^2$. A maximizing vector in $\operatorname{ran} P$ gives the reverse bound with error Ct^2. Apply this argument to $-A(t)$ for the minimum and replace E by $-E$ for negative t. If $A_0 = \mu I$, the result follows directly from the norm bound on eigenvalue perturbations. ∎

Remove the weight phases and the global rotation, and write $G(t) = L^{-1}(WV(t))(WV(t))^*$ for the normalized Gram matrix. Then

$$\begin{aligned} &G(t) = \operatorname{diag}(v_n^2) + tE + t^2F + O(|t|^3), \qquad v_n = |w_n|, \\ &E_{nm} = \mathrm{i}(n-m)v_n v_m \widehat{h}_{n-m}, \qquad F_{nm} = -\tfrac{1}{2}(n-m)^2 v_n v_m \widehat{h^2}_{n-m}. \end{aligned} \tag{3.8}$$

Here $\widehat{h^2}_r = L^{-1}\sum_k h_k^2 \mathrm{e}^{2\pi \mathrm{i} r k/L}$. For $b > a$, let $J_\mu = \{n : v_n^2 = \mu\}$ for $\mu \in \{a^2, b^2\}$. Under the flatness condition $E_{J_\mu J_\mu} = 0$, eliminating the first-order coupling to the complementary space gives the second-order effective matrix (equivalently, the second-order term of the Schur reduction)

$$K_\mu = F_{J_\mu J_\mu} + \left[\sum_{q \notin J_\mu} \frac{E_{nq}E_{qm}}{\mu - v_q^2}\right]_{n,m\in J_\mu}. \tag{3.9}$$

The denominators are nonzero; when $E_{J_\mu J_\mu} = 0$, these are the effective second-order matrices for the two extreme spectral clusters.

THEOREM 3.6 (Necessary and sufficient condition for second-order directions). *As $t \downarrow 0$,*

$$\frac{\kappa_2(WV(t))}{\rho} = 1 + c_+(h)t + O(t^2), \qquad c_+(h) = \frac{\lambda_{\max}(H_{J_+}(h)) - \lambda_{\min}(H_{J_-}(h))}{2}. \tag{3.10}$$

As $t \uparrow 0$, the first-order term is replaced by $c_+(-h)|t|$. The following two conditions are equivalent:

$$\kappa_2(WV(t)) = \rho + O(t^2) \quad (t \to 0); \tag{3.11}$$

$$\widehat{h}_{n-m} = 0 \quad \textit{for all distinct } n, m \textit{ belonging to the same one of } J_-, J_+. \tag{3.12}$$

If $b > a$ and h is not a constant vector, then (3.12) *implies $\kappa_2(WV(t))/\rho - 1 = \Theta(t^2)$, and otherwise $\Theta(|t|)$; here Θ means positive two-sided constant bounds for all sufficiently small nonzero t. If $a = b$, the Fourier condition forces h to be constant, and every nonconstant direction has $\Theta(|t|)$ growth. If h is a constant vector, the nodes only rotate globally and the condition number is identically ρ. If $b > a$ and* (3.12) *holds, then, as $t \to 0$ from either side,*

$$\Delta(t) := \frac{\kappa_2(WV(t))}{\rho} - 1 = c_2(h)t^2 + O(|t|^3), \qquad c_2(h) = \frac{1}{2}\left(\frac{\lambda_{\max}(K_{b^2})}{b^2} - \frac{\lambda_{\min}(K_{a^2})}{a^2}\right). \tag{3.13}$$

For nonconstant h, put

$$C_* := \max_{j \in J_-} C_j, \qquad C_j := \sum_{n \neq j} v_n^2 (n-j)^2 |\widehat{h}_{n-j}|^2.$$

Then

$$c_2(h) \geq \frac{1}{2}\left(\max_{i \in J_+} \frac{(K_{b^2})_{ii}}{b^2} - \min_{j \in J_-} \frac{(K_{a^2})_{jj}}{a^2}\right) \geq \frac{C_*}{2(b^2 - a^2)} > 0. \tag{3.14}$$

In particular, for all sufficiently small nonzero t,

$$\Delta(t) \geq \frac{C_*}{4(b^2 - a^2)} t^2 > 0. \tag{3.15}$$

The factor $1/2$ in (3.14) *bounds the limiting coefficient; the factor $1/4$ in* (3.15) *allows for the remainder at nonzero t.*

Proof. Use the normalized Gram expansion (3.8), with phases removed as above. Lemma 3.5 gives, for $t > 0$,

$$\lambda_{\max}(G(t)) = b^2 + tb^2 \lambda_{\max}(H_{J_+}(h)) + O(t^2),$$
$$\lambda_{\min}(G(t)) = a^2 + ta^2 \lambda_{\min}(H_{J_-}(h)) + O(t^2).$$

The matrix entries are analytic in t, so their remainders, and hence the operator-norm remainder for fixed L, W, h, are bounded by a constant times t^2. Taking the square root of the ratio gives (3.10), including $a = b$ by the scalar case of the lemma.

Since both H_J have zero trace, $\lambda_{\max}(H_{J_+}) \geq 0$ and $\lambda_{\min}(H_{J_-}) \leq 0$. Hence $c_+(h) = 0$ if and only if both compressed matrices vanish, which is exactly (3.12). This condition is unchanged when h is replaced by $-h$, since $H_J(-h) = -H_J(h)$, which gives the two-sided (3.11). If (3.12) fails, the first-order coefficient is strictly positive in both directions.

Suppose now $b > a$ and (3.12) holds. For either extreme eigenvalue μ, the first-order block $E_{J_\mu J_\mu}$ vanishes. Choose a skew-Hermitian matrix S with $S_{nq} = E_{nq}/(v_q^2 - \mu)$ for $n \in J_\mu$, $q \notin J_\mu$, with the opposite block fixed by $S^* = -S$ and zero diagonal blocks. In $\mathrm{e}^{-tS}G(t)\mathrm{e}^{tS}$, the first-order off-diagonal block vanishes and the J_μ block is $\mu I + t^2 K_\mu + O(|t|^3)$. A second unitary transformation of size $O(t^2)$ removes the remaining second-order off-diagonal block without changing this diagonal block through order two. The complementary spectral gap is fixed, so both transformations have bounded coefficients. The Hermitian eigenvalue norm bound yields $\mu + t^2\lambda_{\mathrm{ext}}(K_\mu) + O(|t|^3)$ on both sides of zero. Taking the square root of the extreme-eigenvalue ratio proves (3.13).

For a direct Schur-complement check, write $J = J_\mu$ and let Q be its complement. For an eigenvalue $\lambda = \mu + O(t^2)$ in this cluster, the complementary block equation gives

$$\lambda u = \big[G_{JJ} + G_{JQ}(\lambda I - G_{QQ})^{-1} G_{QJ}\big] u.$$

The fixed gap implies $(\lambda I - G_{QQ})^{-1} = (\mu I - \mathrm{diag}(v_q^2)_{q\in Q})^{-1} + O(t)$. Since both coupling blocks are $O(t)$, the reduced matrix is $\mu I + t^2 K_\mu + O(|t|^3)$, with the plus sign and denominators in (3.9).

The diagonal entries of F vanish, so (3.9) gives

$$\frac{(K_{a^2})_{jj}}{a^2} = -\sum_{q\notin J_-} \frac{v_q^2 (j-q)^2 |\widehat{h}_{j-q}|^2}{v_q^2 - a^2}, \qquad j \in J_-,$$

$$\frac{(K_{b^2})_{ii}}{b^2} = \sum_{q\notin J_+} \frac{v_q^2 (i-q)^2 |\widehat{h}_{i-q}|^2}{b^2 - v_q^2}, \qquad i \in J_+.$$

The top diagonal entries are nonnegative. In the bottom formula, $0 < v_q^2 - a^2 \leq b^2 - a^2$, and flatness eliminates the terms with $q \in J_- \setminus \{j\}$ from C_j. Rayleigh bounds for the diagonal entries therefore give (3.14). For fixed j, the differences $n - j$, $0 \leq n < L$, form a complete residue system modulo L; a nonconstant real h has a nonzero Fourier coefficient at a nonzero residue, so $C_j > 0$. The expansion (3.13) then implies (3.15).

The moment constraint gives the same coarser coefficient bound independently: $s_r(t) = \mathrm{i}r\widehat{h}_r t + O(t^2)$ implies $T_j = C_j t^2 + O(|t|^3)$, while $\eta = 2\Delta + \Delta^2$. Thus Proposition 3.4 gives $C_j \leq 2(b^2 - a^2)c_2(h)$ after division by t^2 and passage to the limit. This also explains the role of that constraint in proving positivity. If $a = b$, both extreme index sets are the whole space, so (3.12) eliminates every nonzero Fourier coefficient and forces h to be constant. Finally, a constant h changes $WV(t)$ only by a unitary diagonal factor on the left. ∎

A non-triangular example with repeated extreme weights.. Take $L = 4$, $W = \mathrm{diag}(1, 2, 1, 2)$ and $h = (1, 0, -1, 0)$. Here $J_- = \{0, 2\}$, $J_+ = \{1, 3\}$ and $\widehat{h}_2 = 0$, so the Fourier flatness condition holds. In these

respective index orders,

$$K_1 = \frac{1}{3}\begin{bmatrix} -10 & -5 \\ -5 & -2 \end{bmatrix}, \qquad K_4 = \frac{1}{3}\begin{bmatrix} 2 & -10 \\ -10 & 10 \end{bmatrix}.$$

Their relevant eigenvalues are $\lambda_{\min}(K_1) = -2 - \sqrt{41}/3$ and $\lambda_{\max}(K_4) = 2 + 2\sqrt{29}/3$. Hence

$$\frac{\kappa_2(WV(t))}{2} - 1 = \left(\frac{5}{4} + \frac{\sqrt{29}}{12} + \frac{\sqrt{41}}{6}\right) t^2 + O(|t|^3).$$

The coefficient is approximately 2.765951; its diagonal and moment lower bounds in (3.14) are 25/12 and 5/3, respectively. This even-dimensional example exhibits coupling in both extreme effective blocks without the triangular weights of the polynomial application.

COROLLARY 3.7 (First-harmonic criterion for the triangular weights). *For $L = 2d+1$, $W = D$ and the real perturbation directions of* (3.7)*,*

$$\frac{\kappa_2(DV(t))}{d+1} = 1 + d|\widehat{h}_1|\,|t| + O(t^2). \tag{3.16}$$

Hence $\kappa_2(DV(t)) = d + 1 + O(t^2)$ holds if and only if

$$\sum_{k=0}^{2d} h_k \mathrm{e}^{2\pi \mathrm{i} k/(2d+1)} = 0. \tag{3.17}$$

After excluding global rotations, the directions satisfying this condition have exactly second-order growth, and all other directions grow first-order.

Proof. Here $J_- = \{0, 2d\}$ and $J_+ = \{d\}$. The compressed matrix for the latter is zero, and the former has the two eigenvalues $\pm 2d|\widehat{h}_{2d}| = \pm 2d|\widehat{h}_1|$. Theorem 3.6 applies. ∎

Condition (3.17) consists of two real linear constraints; the solution space has dimension $L-2$, or $L-3$ after removing global rotations, and the Euclidean orthogonal projection of an arbitrary direction onto it is

$$h_k^{\mathrm{flat}} = h_k - 2\,\mathrm{Re}\big(\widehat{h}_1 \mathrm{e}^{-2\pi \mathrm{i} k/L}\big). \tag{3.18}$$

For example, when $d \geq 2$, $h_k = \cos(2\pi k/L)$ gives the first-order coefficient $d/2$, while $h_k = \cos(4\pi k/L)$ satisfies the criterion with strictly second-order growth. If the largest and smallest moduli are each attained only once, (3.12) has no nontrivial index pairs and every non-rotational direction is second-order; if all weights have equal modulus, the condition forces h to be constant.

PROPOSITION 3.8 (An explicit second-harmonic coefficient). *For $d \geq 2$, $L = 2d+1$ and $h_k = \cos(4\pi k/L)$,*

$$\frac{\kappa_2(DV(t))}{d+1} = 1 + C_d t^2 + O(|t|^3), \qquad C_d = \begin{cases} 21/8, & d = 2, \\ \dfrac{(d-1)^2}{4d} + \dfrac{9}{16} + \dfrac{(2d-1)^2}{6}, & d \geq 3. \end{cases} \tag{3.19}$$

Proof. Apply Theorem 3.6 with $W = D$. Here $\widehat{h}_r = 1/2$ for $r \equiv \pm 2 \pmod{L}$ and is zero otherwise, while $\widehat{h^2}_r = 1/2$ for $r \equiv 0$ and $1/4$ for $r \equiv \pm 4$. For $b = d+1$, the top block is scalar and $K_{b^2}/b^2 = (d-1)^2/(2d)$, from the indices $d \pm 2$. For the bottom block, ordered as $(0, 2d)$, $F_{00} = F_{2d,2d} = 0$. The only intermediate indices contributing to $(K_1)_{00}$ are $q = 2, 2d-1$, which give

$$(K_1)_{00} = \frac{9}{1-9} + \frac{(2d-1)^2}{1-4} = -\frac{9}{8} - \frac{(2d-1)^2}{3} = (K_1)_{2d,2d}.$$

For $d \geq 3$, $F_{0,2d} = 0$ and the two bottom indices have no common intermediate index, so the off-diagonal entry vanishes. For $d = 2$, the Fourier alias modulo 5 gives $F_{0,4} = -2$, and the common intermediate index $q = 2$ adds $9/8$. Thus

$$K_1 = \begin{cases} \begin{bmatrix} -33/8 & -7/8 \\ -7/8 & -33/8 \end{bmatrix}, & d = 2, \\ -\left(\dfrac{9}{8} + \dfrac{(2d-1)^2}{3}\right) I_2, & d \geq 3. \end{cases} \tag{3.20}$$

Consequently (3.20) gives $\lambda_{\min}(K_1) = -5$ for $d = 2$, and equals the diagonal entry otherwise. Taking the square root of the extreme-eigenvalue ratio gives $C_d = \frac{1}{2}(K_{b^2}/b^2 - \lambda_{\min}(K_1))$. ∎

Remark 3.9 (Dimension dependence). For $d \geq 3$, the coefficient in (3.19) is

$$C_d = \frac{2}{3}d^2 - \frac{5}{12}d + \frac{11}{48} + \frac{1}{4d}, \qquad C_d \sim \frac{2}{3}d^2 \quad (d \to \infty). \tag{3.21}$$

The directions $h_k = \cos(4\pi k/(2d+1))$ satisfy $\max_k |h_k| = 1$, so $|t|$ is their maximum angular displacement. Even after eliminating the first harmonic, the second-order sensitivity in this normalization grows quadratically with d. The asymptotic expansion holds as $t \to 0$ for each fixed d; the behavior of C_d is then examined as d increases.

4. Explicit Fourier formulas and error propagation.

4.1. Weight computation and a weighted Parseval identity. Let $\zeta = \mathrm{e}^{2\pi \mathrm{i}/L}$, and first take $\theta = 0$ with the optimal nodes in their natural order. Since $V^{-1} = V^*/L$, the unique weights are

$$\lambda_k = \frac{1}{L}\sum_{n=0}^{m} \frac{A_n}{N_n}\zeta^{-kn}, \qquad k = 0, \ldots, L-1. \tag{4.1}$$

A general rotation θ only requires replacing the normalized coefficients by $a_n \mathrm{e}^{-\mathrm{i}n\theta}$ in the same formula.

Proposition 4.1 (Weight norms and input perturbations). *At the optimal nodes,*

$$\|\lambda\|_2^2 = \frac{1}{L}\sum_{n=0}^{m} \frac{|A_n|^2}{N_n^2}. \tag{4.2}$$

If the nodes are fixed and only the input coefficients are changed to $A + \delta A$, then the exact solution changes by

$$\|\delta\lambda\|_2^2 = \frac{1}{L}\sum_{n=0}^{m} \frac{|\delta A_n|^2}{N_n^2}. \tag{4.3}$$

Moreover, for $A \neq 0$,

$$\frac{\|\delta\lambda\|_2}{\|\lambda\|_2} \leq (d+1)\frac{\|\delta A\|_2}{\|A\|_2}, \tag{4.4}$$

and the constant $d+1$ is attained.

Proof. Identity (4.2) follows from $\lambda = V^* D^{-1} A/L$ and $VV^* = LI$; applying the same identity to δA gives (4.3). Since $1 \leq N_n \leq d+1$,

$$\|\delta\lambda\|_2 \leq \frac{\|\delta A\|_2}{\sqrt{L}}, \qquad \|\lambda\|_2 \geq \frac{\|A\|_2}{\sqrt{L}(d+1)},$$

which proves (4.4). Taking $A = e_d$ and $\delta A = \varepsilon e_0$ gives

$$\frac{\|\delta A\|_2}{\|A\|_2} = |\varepsilon|, \qquad \frac{\|\delta\lambda\|_2}{\|\lambda\|_2} = (d+1)|\varepsilon|,$$

so the uniform relative constant cannot be reduced. ∎

Identity (4.3) also shows that equal absolute errors on different monomial coefficients affect the weights differently. With the weighted input norm $\|A\|_{D^{-1}} := \left\|D^{-1}A\right\|_2$, the weight map differs from an isometry only by the constant $1/\sqrt{L}$, and the corresponding relative perturbation bound has constant 1.

4.2. Exact input perturbations versus implementation errors. Proposition 4.1 describes the exact linear transformation; floating-point errors are determined jointly by the preprocessing and the FFT computation. Schatzman [11] studied the accuracy of the DFT and the FFT, including twiddle-factor errors, and Higham [6] the errors of floating-point summation schemes, indicating that the weight transformation and the explicit coefficient expansion should be examined along separate paths. Suppose the preprocessing produces $\widehat{a} = a + e_a$ with $\|e_a\|_2 \le \eta_a \|a\|_2$, and write $U = V^*/L$; assume that the result of the FFT and the final scaling satisfies

$$\widehat{\lambda} = U\widehat{a} + e_U, \qquad \|e_U\|_2 \le \eta_U \|U\widehat{a}\|_2 .$$

Since U is a constant multiple of a unitary matrix, the conditional estimate

$$\frac{\left\|\widehat{\lambda} - \lambda\right\|_2}{\|\lambda\|_2} \le \eta_a + \eta_U(1+\eta_a) \tag{4.5}$$

holds, where η_U represents the implementation error of the concrete FFT library; below, the accuracy of the actual computation is assessed against an independent high-precision DFT.

If, in addition, the matrix incurs an error E and the computed result satisfies $(M+E)\widehat{\lambda} = A + e$ exactly with $q = \left\|M^{-1}\right\|_2 \|E\|_2 < 1$, then the standard perturbation estimate for linear systems [7, Chapter 7] gives

$$\frac{\left\|\widehat{\lambda} - \lambda\right\|_2}{\|\lambda\|_2} \le \frac{\kappa_2(M)}{1-q}\left(\frac{\|e\|_2}{\|A\|_2} + \frac{\|E\|_2}{\|M\|_2}\right). \tag{4.6}$$

Storing the ideal unit roots as floating-point numbers and expanding the squares explicitly thus introduces an error source distinct from the FFT weight computation; the numerical section reports these quantities separately.

4.3. Bounded angular node perturbations: an application of the discrete Kadec bound. Consider

$$\widetilde{\beta}_k = \mathrm{e}^{\mathrm{i}(\theta + 2\pi k/L + \delta_k)}, \qquad |\delta_k| \le \frac{\alpha\pi}{L}, \quad L = 2d+1.$$

The angular spacing of adjacent ideal nodes is $\Delta_\theta = 2\pi/L$, so the perturbation occupies the fraction $\alpha/2$ of the spacing.

Corollary 4.2 (Weighted condition number under perturbed nodes). *For $L = 2d+1$, if $0 \le \alpha < 1/2$ and*

$$\varphi(\alpha) = 1 - \cos(\pi\alpha/2) + \sin(\pi\alpha/2) < 1,$$

then for arbitrary nonzero fixed weights W and $\widetilde{V} = (\widetilde{\beta}_k^n)$,

$$\rho \le \kappa_2(W\widetilde{V}) \le \rho \frac{1+\varphi(\alpha)}{1-\varphi(\alpha)}. \tag{4.7}$$

For the geometric square, $\rho = d+1$.

Proof. Shifting the power indices from $0, \dots, 2d$ to $-d, \dots, d$ amounts to right multiplication by the unitary diagonal matrix $\operatorname{diag}(\widetilde{\beta}_k^{-d})$ and does not change the singular values; the rotation θ also contributes only a unitary diagonal factor. Applying Yu and Townsend [13, Theorem 2.1, Corollary 2.3] with perturbation fraction $\alpha/2 < 1/4$ gives

$$\sigma_{\min}(\widetilde{V}) \ge \sqrt{L}(1-\varphi), \qquad \sigma_{\max}(\widetilde{V}) \le \sqrt{L}(1+\varphi).$$

The product inequality for condition numbers together with Theorem 3.1 completes the proof. The perturbation range keeps the nodes distinct. ∎

For fixed $\alpha < 1/2$, (4.7) controls the amplification relative to ρ by a factor independent of L. At $\alpha = 1/2$, the upper bound becomes uninformative. The numerical experiments report endpoint behavior separately.

5. Algorithm, batch computation and an exact low-degree example.

5.1. Workflow and cost.

Given d and the input coefficients A, the fixed-node workflow is as follows:

1. set $L = 2d+1$ and compute the integer weights $N_n = \min(n+1, L-n)$;
2. compute $a_n = A_n/N_n$;
3. compute the DFT of length L with the negative-exponent convention and set $\lambda = \mathrm{FFT}(a)/L$;
4. represent the square terms through the shared nodes $\beta_k = \mathrm{e}^{2\pi \mathrm{i} k/L}$ and (1.2);
5. verify, as needed, the coefficient reconstruction or an independent high-precision weight error.

With a suitable FFT algorithm, computing all weights requires $O(L \log L)$ arithmetic operations and the normalization requires $O(L)$. For B homogeneous polynomials of the same degree, arrange the input coefficients as an $L \times B$ matrix and perform the FFT along the first axis, at a cost of $O(BL \log L)$.

For comparison, the precomputed dense operator $R = V^* D^{-1}/L$ applies in $O(BL^2)$ with $O(L^2)$ storage, while a general dense solver costs $O(L^3)$ per factorization plus $O(BL^2)$ for a batch of B right-hand sides. Caching its LU factors avoids repeating the factorization. The experiments record preparation times separately and exclude them from the online timings; at small sizes, library call overheads and optimized matrix multiplication may change the actual time ordering.

The FFT complexity refers only to the weights and the shared-node representation; explicitly outputting all L squares with their $d+1$ coefficients each is an $O(d^2)$ task and cannot be counted as $O(d \log d)$.

5.2. Conjugate pairing for real input and signed squares.

Proposition 5.1 (Real signed-square representation). *Let $A \in \mathbb{R}^{2d+1}$ and take the unrotated unit roots $\beta_k = \zeta^k$. Then $\lambda_0 \in \mathbb{R}$ and $\lambda_{L-k} = \overline{\lambda_k}$. For $1 \le k \le d$, choose any $c_k \in \mathbb{C}$ with $c_k^2 = \lambda_k$, and set*

$$U_k = \operatorname{Re}(c_k Q_{\zeta^k}), \qquad T_k = \operatorname{Im}(c_k Q_{\zeta^k}),$$

where the real and imaginary parts are taken coefficientwise. Then

$$F = \lambda_0 Q_1^2 + 2\sum_{k=1}^{d}(U_k^2 - T_k^2). \tag{5.1}$$

Consequently, F is a real linear combination, with negative coefficients allowed, of at most $2d+1$ squares of real homogeneous forms of degree d.

Proof. Conjugation of a polynomial is taken coefficientwise. The DFT of the real vector $D^{-1}A$ has the stated conjugate symmetry, and $Q_{\zeta^{L-k}} = \overline{Q_{\zeta^k}}$. Each pair contributes

$$\lambda_k Q_{\zeta^k}^2 + \overline{\lambda_k Q_{\zeta^k}^2} = 2\,\mathrm{Re}\big[(U_k + \mathrm{i}T_k)^2\big] = 2(U_k^2 - T_k^2),$$

and the term at the real node $\beta_0 = 1$ is added. The two choices of c_k only flip the signs of U_k and T_k simultaneously and leave the square representation unchanged; if $\lambda_k = 0$, take $c_k = 0$. ∎

In implementation, only the $d+1$ outputs of $\mathrm{rfft}(D^{-1}A)/L$ are needed, the rest being completed by conjugate symmetry. To make the coefficients ± 1, absorb $\sqrt{|\lambda_0|}$ and $\sqrt{2}$ into the real forms and omit zero terms. The construction allows negative signs and is therefore not a nonnegative sum-of-squares certificate; U_k and T_k in general leave the original one-parameter geometric family, and explicitly outputting the real forms still costs $O(d^2)$ coefficients.

5.3. An exact decomposition at fixed unit-root nodes. Take $d = 1$, $L = 3$ and $F = x^2 + y^2$, so that $N = (1, 2, 1)$ and $a = (1, 0, 1)$. Let $\zeta = \mathrm{e}^{2\pi\mathrm{i}/3}$. Formula (4.1) gives

$$\lambda_0 = \frac{2}{3}, \qquad \lambda_1 = \frac{1+\zeta}{3}, \qquad \lambda_2 = \frac{1+\zeta^2}{3},$$

and hence

$$x^2 + y^2 = \frac{2}{3}(x+y)^2 + \frac{1+\zeta}{3}(x+\zeta y)^2 + \frac{1+\zeta^2}{3}(x+\zeta^2 y)^2. \tag{5.2}$$

Computing the coefficients of x^2, xy and y^2 on the right-hand side and using $1 + \zeta + \zeta^2 = 0$ gives 1, 0 and 1, respectively. The example illustrates the complex weighted representation at fixed nodes and makes no claim of a minimal number of terms.

6. Numerical experiments.

6.1. Environment, data and structural checks.

Computational environment.. The reported experiments were obtained from the complete run of 28 October 2026 on a Lenovo 12S9CTO1WW computer with an Intel Core i5-13500 processor (reported base frequency 2.50 GHz) and 8 GB installed RAM (7.66 GiB reported by the operating system). The software environment was Windows 11 (build 26200), 64-bit Python 3.12.10, NumPy 2.5.4, SciPy 1.18.1, Matplotlib 3.11.2, mpmath 1.4.1 and threadpoolctl 3.7.0. The NumPy and SciPy BLAS backends were OpenBLAS 0.3.34.106.0 and 0.3.31.dev, respectively, both limited to one thread. The random seeds are 20261023 for the computational accuracy, node-family and input-perturbation experiments and 20261024 for the structural checks and batch benchmarks. The local-direction experiments are deterministic: eigenvalues and Fourier moments are computed at 80 decimal digits, with selected tiny perturbations rechecked at 120 digits. Programs, raw data and environment records accompany the article in `../code`; the hardware details supplied after the run are identified separately in its hardware supplement.

Reproducibility.. This complete run reproduced all non-timing CSV records of the earlier Python 3.12.14 run exactly, as checked by rowwise comparison or SHA-256 hashes. Table 2, Figure 3 and the timing values below use the 28 October run, together with its raw repetitions and environment records. Earlier records are archived separately in `../reproducibility`. The code README gives commands for isolated reruns and regeneration of all three figures from the adopted CSVs; fresh timings may vary with the execution environment.

The structural checks comprise 130 general-weight configurations ($L = 4, 5, 8, 9, 16, 17$, minimum weight in every row, complex phase and equal-modulus weights) and 240 triangular-weight node perturbations, of which 200 satisfy the strict range of Corollary 4.2 and 40 are endpoint observations at $\alpha = 1/2$; all pass the applicable checks. The real-square construction is verified on 64 coefficient-convolution reconstructions with maximum normalized error about 6.10×10^{-14}. The remaining node-family, input-perturbation and higher-power checks are kept in the reproduction materials.

6.2. Local growth orders and the moment constraint. For Theorem 3.6 we examine 45 weight-direction configurations: the triangular weights, weights with unique extreme moduli, weights with repeated extreme moduli and equal-modulus weights. The directions include global rotations, several real Fourier modes and the projections (3.18). For each we take $t = \pm 10^{-2}, \pm 10^{-4}, \pm 10^{-6}, \pm 10^{-8}$, giving 360 matrices, and verify the left and right first-order coefficients and the growth order of non-rotational directions.

Figure 1 illustrates triangular weights with $L = 17$. The first-harmonic direction $h_k = \cos(2\pi k/L)$ approaches $4|t|$, consistent with $d/2 = 4$ in Corollary 3.7, while the second harmonic and the projected directions (started from $h_k = (-1)^k$) exhibit second-order growth. The script also checks negative perturbations, complex weight phases and changes of the extreme multiplicities.

For the general coefficient in (3.13), the 22 nonconstant flat directions in this sweep are checked at $t = \pm 10^{-5}, \pm 10^{-7}$, using 88 independent full-matrix eigenvalue computations at 80 digits. The effective coefficient, the diagonal lower bound and the moment lower bound are compared; two selected cases are replayed at 120 digits. The $L = 4$ repeated-extreme example is additionally checked using exact rational effective blocks and 12 full-matrix calculations at 80 or 120 digits. Exact rational checks cover both branches of (3.5), including equal weight moduli.

For Proposition 3.8, exact rational evaluation of (3.9) verifies the displayed coefficient for $2 \le d \le 64$. Independent full-matrix eigenvalue computations for $d = 2, 3, 4, 5, 8, 12$ and $t = \pm 10^{-4}, \pm 10^{-6}, \pm 10^{-8}$, together with six 120-digit checks at $t = \pm 10^{-10}$, verify convergence to C_d. For example, $C_2 = 21/8$ and $C_8 = 1267/32$; the largest coefficient error at $|t| = 10^{-8}$ is 6.33×10^{-9} over the tested degrees.

To assess the looseness of the upper bound of Proposition 3.4, we take $L = 5, 9, 17$ with two families of positive weights: the triangular weights; and the interior-minimum family $w_{\lfloor L/2 \rfloor} = 1$ with $w_n = 2 + n/L$ for the remaining rows, whose largest and smallest weights are both attained once. For notational simplicity we write its minimum row as $j = \lfloor L/2 \rfloor$; the triangular weights take $j = 0$. The nodes are uniformly

$$\beta_k(\varepsilon) = \exp\left(\frac{2\pi \mathrm{i}}{L}(k + \varepsilon q_k)\right), \qquad q_k = \frac{L(-1)^k - 1}{L + 1}.$$

Here $\sum_k q_k = 0$ and $\max_k |q_k| = 1$, so ε is the largest perturbation fraction relative to the grid spacing. We take

$$\varepsilon \in \{10^{-12}, 10^{-10}, 10^{-8}, 10^{-6}, 10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}\},$$

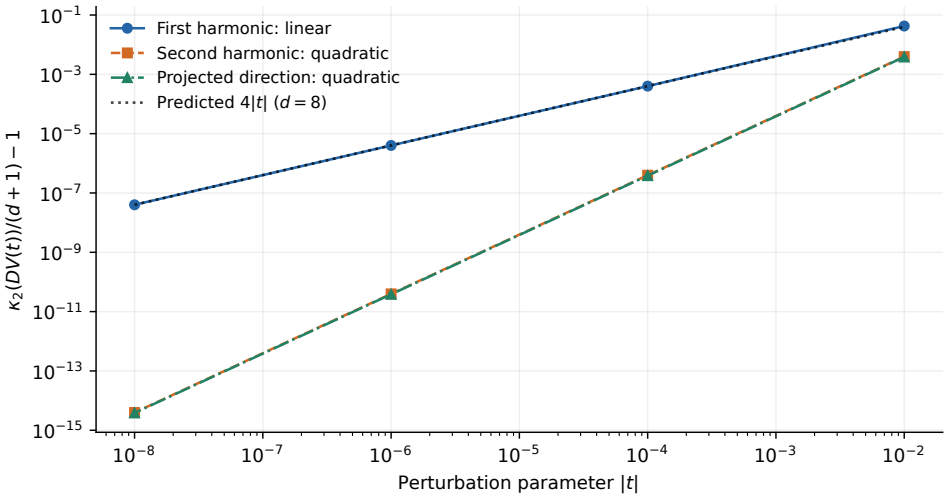


FIGURE 1. *Local condition-number increments under the triangular weights with $d = 8$. First-order growth when the first harmonic is present; second-order growth for non-rotational directions with vanishing first harmonic. The figure uses the 80-digit computations.*

giving 48 cases, and additionally verify 6 equality configurations with $\varepsilon = 0$.

Write T_j and $H_{a,b}(\eta)$ for the left- and right-hand sides of (3.5), and report $T_j/H_{a,b}(\eta)$ for the nonzero perturbations. All cases satisfy the upper bound. For the interior-minimum weights the ratio lies between about 0.250 and 0.334; for the triangular weights the ratio decreases markedly as the perturbation shrinks, with a smallest measured value of about 4.15×10^{-12}. The observed differences in bound tightness reflect the distinct extreme-weight multiplicities of the two families.

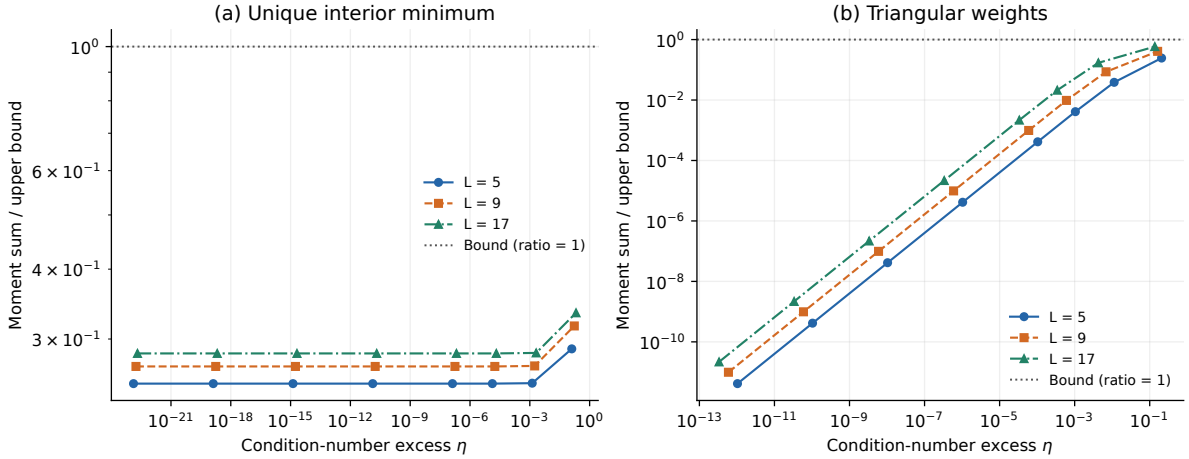


FIGURE 2. *Ratio of the left-hand side of the near-optimal moment constraint to its upper bound. Each curve corresponds to one fixed degree and a deterministic direction; the dashed line marks ratio* 1. *The two weight families illustrate different bound tightness associated with their extreme-weight multiplicities.*

The difference is consistent with Theorem 3.6: in the interior-minimum example both extremes are

attained once and η is second-order for non-rotational directions, whereas for the triangular weights the present direction has a nonzero first harmonic and η is first-order. In both families T_j is second-order, so the ratio can retain a nonvanishing magnitude in the former and tends to zero in the latter.

6.3. Computational accuracy and batch efficiency. Define the weight forward error and the coefficient reconstruction residuals

$$E_\lambda = \frac{\left\|\widehat{\lambda} - \lambda_{\mathrm{ref}}\right\|_2}{\|\lambda_{\mathrm{ref}}\|_2}, \qquad R_{\mathrm{dense}} = \frac{\left\|\widehat{M}\widehat{\lambda} - A\right\|_2}{\|A\|_2}, \qquad R_{\mathrm{fft}} = \frac{\left\|D(L\,\mathrm{IFFT}(\widehat{\lambda})) - A\right\|_2}{\|A\|_2}. \tag{6.1}$$

Here $(\mathrm{IFFT}\, z)_n = L^{-1}\sum_{k=0}^{L-1} z_k \mathrm{e}^{2\pi \mathrm{i} nk/L}$, and λ_{ref} is obtained by direct DFT summation at 80-digit precision, and $\widehat{M}$ is constructed independently from floating-point nodes and successive powers. FFT round trips may share numerical structure, so R_{fft} alone is not used to certify the weight accuracy. We take $d = 2, 8, 32, 128$ and generate three instances each of random complex coefficients, sparse coefficients with at most 4 nonzero entries, and wide-scale coefficients whose moduli cover 10^{-8} to 10^{8}, for 36 cases in total.

TABLE 1
Largest errors over the nine cases of each degree, in e-notation.

d	L	E_λ	R_{dense}	R_{fft}
2	5	1.40e-16	6.31e-16	2.93e-16
8	17	1.74e-16	1.53e-14	6.43e-16
32	65	2.47e-16	4.85e-14	1.07e-15
128	257	5.17e-16	2.61e-13	4.96e-15

In Table 1 the largest forward error of the ideal weights is about 5.17×10^{-16} and the largest dense-expansion residual about 2.61×10^{-13}; the explicit node construction, power computation and summation add rounding, so the three metrics need not coincide. A further 20 equality-case and 400 random cases verify Proposition 4.1; in the equality cases the normalized amplification differs from 1 by at most 1.52×10^{-5}, occurring at the smallest perturbations and including the cancellation error of subtracting two floating-point solutions.

The batch experiment takes $d = 8, 32, 128, 512$ and $B = 1, 128$, using identical complex128 inputs for the FFT, a precomputed dense operator, cached LU solves and LU factorization followed by a solve at every call. Both LU methods use the same SciPy routines and options. All methods are warmed up; their order is randomized within each of 7 repetitions, with enough inner calls to reduce timer noise. Figure 3 reports median online times per batch, excluding input generation and fixed-operator setup. The fresh-LU timing includes factorization; the cached-LU timing excludes it. Preparation is reported in Table 2. All solver outputs are also checked against the FFT weights and by dense residuals.

For $d = 512$ and $B = 128$, the FFT, cached dense transform, cached LU and fresh LU take 5.73, 25.1, 28.6 and 111 ms per batch, respectively. At smaller sizes, the cached dense transform can be faster than the FFT; the ordering also varies with batch size. Raw repetitions, setup times, solver residuals and library records are supplied in `../code/ela_revision`; the measurements describe this implementation and platform only.

TABLE 2

Median preparation times in milliseconds. Stages are timed separately: cached LU requires matrix construction and factorization; the dense transform is constructed directly.

d	Weights	Matrix M	Dense operator R	LU of M
8	0.00396	0.0262	0.017	0.00819
32	0.00342	0.315	0.129	0.0507
128	0.00492	6.91	3.58	1.75
512	0.00632	109	63.2	78.2

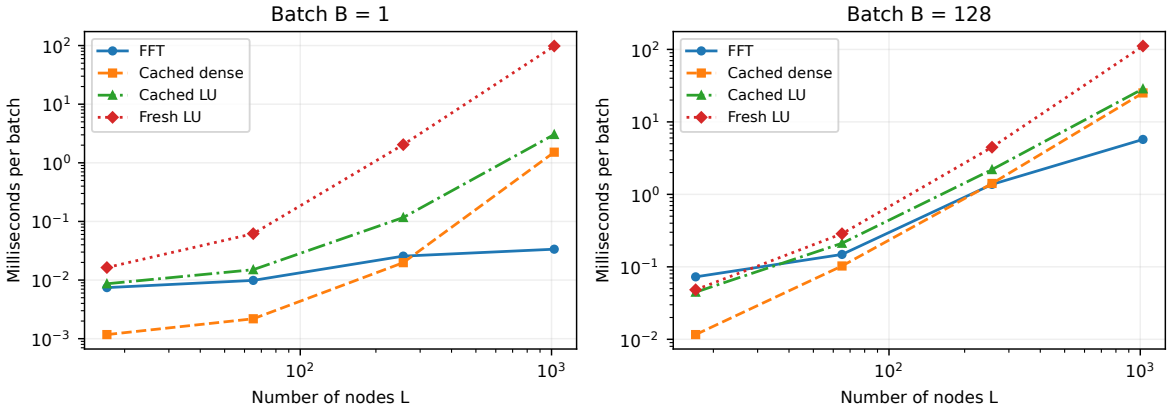


FIGURE 3. *Median online timings per batch call; left panel $B = 1$, right panel $B = 128$. Both cached methods reuse their preparation; fresh LU refactorizes at each call.*

7. Discussion and conclusion. For fixed nonzero monomial weights, the optimal unit-circle Vandermonde configurations are exactly the rotated regular polygons. The local Fourier criterion identifies which directions avoid linear growth, and the weighted moment constraint ensures strictly quadratic growth in every non-rotational flat direction when the weight moduli are unequal. Formula (3.13) computes the coefficient for every such direction, and (3.14) gives positive lower bounds. For triangular weights, one Fourier coefficient decides the order; the second harmonic also admits the explicit coefficient (3.19). By (3.21), $C_d \sim 2d^2/3$ in the maximum-angular-displacement normalization of Remark 3.9; removing linear growth therefore still leaves a dimension-dependent quadratic sensitivity.

The geometric-square application supplies a fixed basis for binary forms. Its FFT formula, Parseval identity and Kadec upper bound use existing tools; the experiments check the local theory and distinguish ideal-weight accuracy from explicit-node reconstruction errors. Cached LU and a precomputed dense transform provide reusable baselines for the batch computation. Timing advantages remain dependent on matrix size, batch size and implementation.

Questions left open include uniform control of higher-order remainders, node-matching bounds from the moment constraint, and cancellation in real square factors.

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